

# Audio & Speech: Audio Reconstruction from MFCC

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DSP Lab 2025 autumn Audio and Speech week 2

授課老師：李祈均教授

助教：莊翔凱、周星航

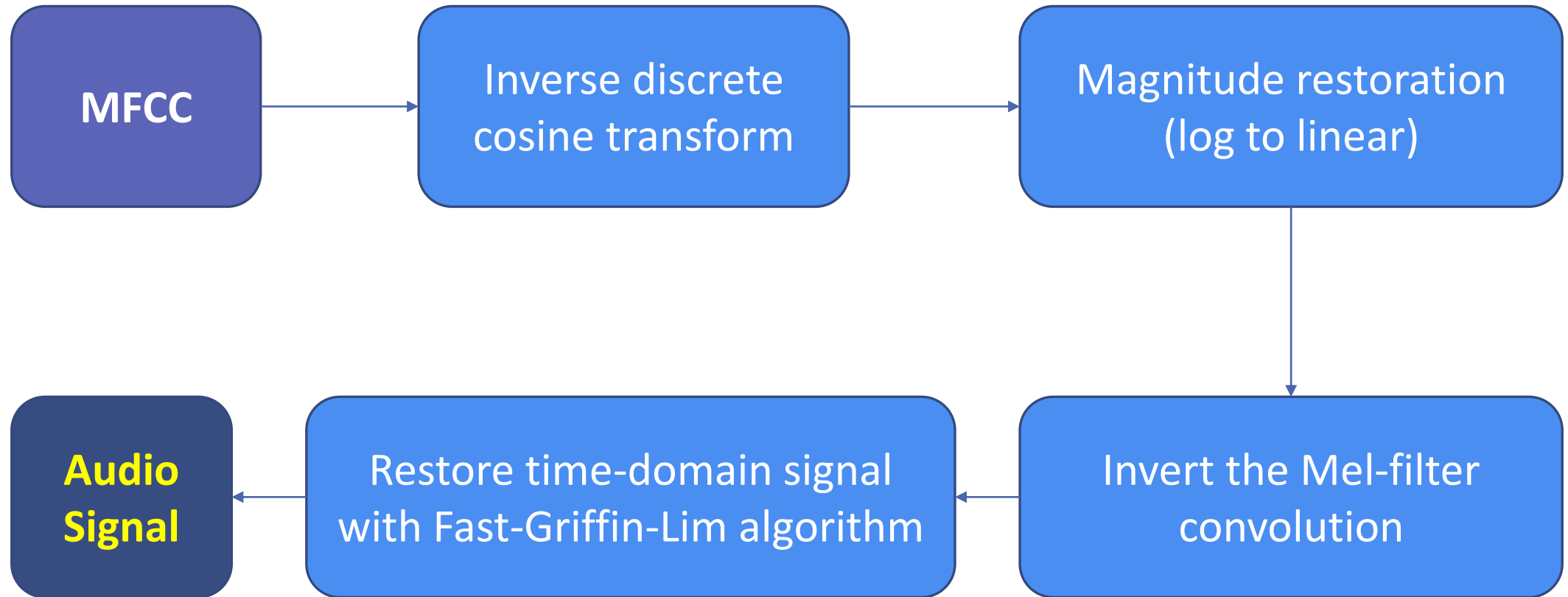
# Outline:

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- MFCC inversion overview.
- Elaboration on some components.
- Demo.
- Report requirements.

# Audio reconstruction flow chart

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# Component(1): Filter inv-convolution

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- Recall that the convolution was:

$$\mathbf{Features} = \mathbf{STFT}\{\mathbf{signal}\} * \mathbf{Mel-Filter}$$

- Notice that for a linear transformation  $\mathbf{Y} = \mathbf{XA}$  with given  $\mathbf{Y}$  and  $\mathbf{A}$ , the  $\mathbf{X}$  that yields ordinary least square:  $(\mathbf{Y} - \mathbf{XA})^T(\mathbf{Y} - \mathbf{XA})$  can be found with equation:

$$\mathbf{X} = (\mathbf{A}^T \mathbf{A})^\dagger \mathbf{A}^T \mathbf{Y}$$

- Where  $\mathbf{Y}$  is *Features*, the *Mel-Filter* is  $\mathbf{A}$ , and we want  $\mathbf{STFT}\{\mathbf{signal}\}$  as  $\mathbf{X}$ .

# Component(2): Fast Griffin-Lim

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$$P_{C_1}(c) = STFT\{ invSTFT\{ c \} \}$$

$$P_{C_2}(c) = s \cdot e^{i\angle c}$$

$\alpha_n$ : step size

$$G^\dagger: invSTFT\{$$

**Fix** the initial phase  $\angle c_0$

**Initialize**  $c_0 = s \cdot e^{i\angle c_0}$ ,  $t_0 = P_{C_2}(P_{C_1}(c_0))$

**Iterate** for  $n = 1, 2, \dots$

$$t_n = P_{C_1}(P_{C_2}(c_{n-1}))$$

$$c_n = t_n + \alpha_n(t_n - t_{n-1})$$

Update  $\alpha_n$

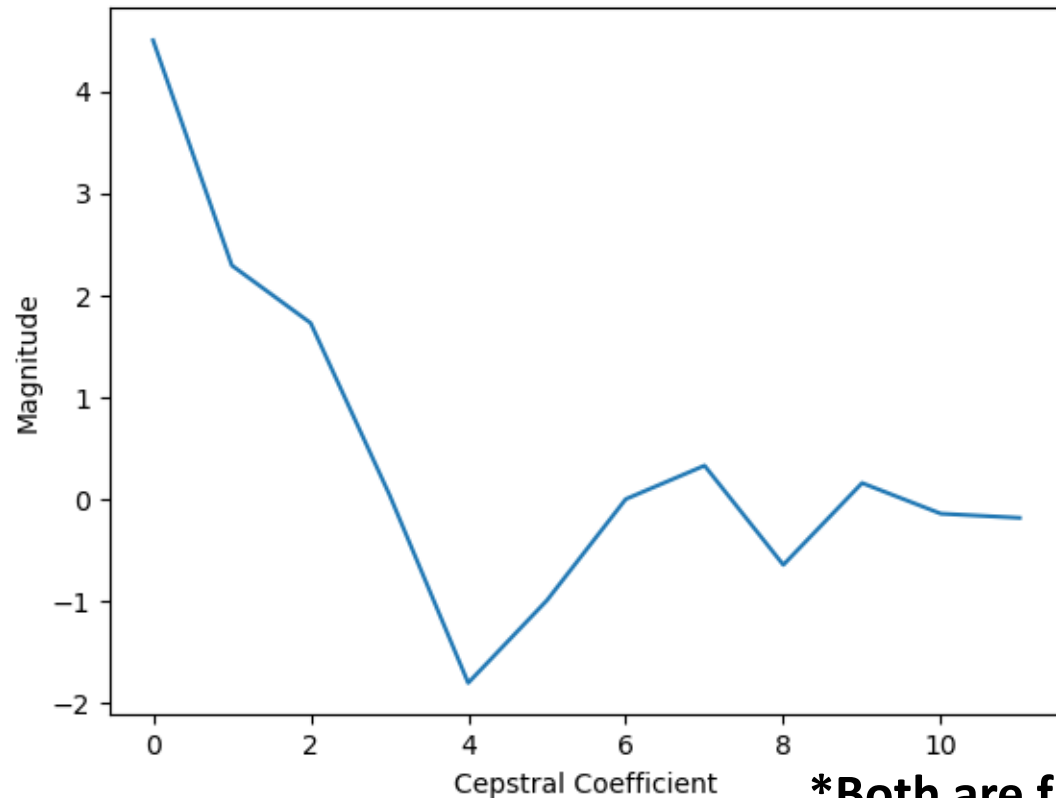
**Until convergence**

$$x^* = G^\dagger c_n$$

# Demo(1): Effect of num Mel-Filter banks

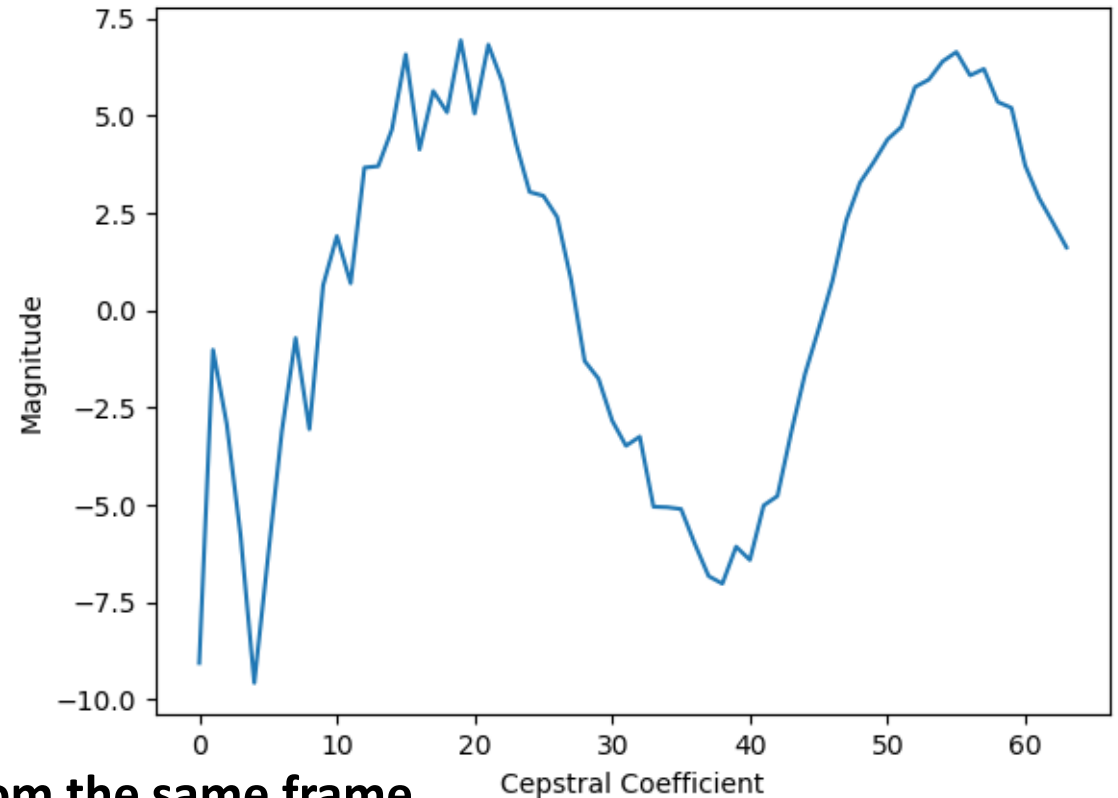
**12 banks**

MFCC of a random frame



**64 banks**

MFCC of a random frame

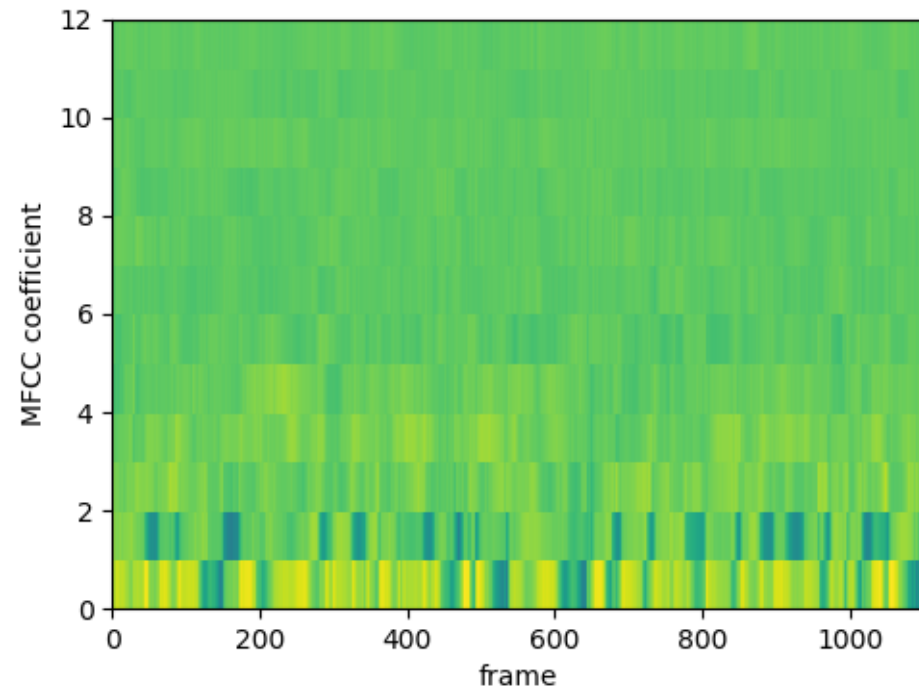


**\*Both are from the same frame**

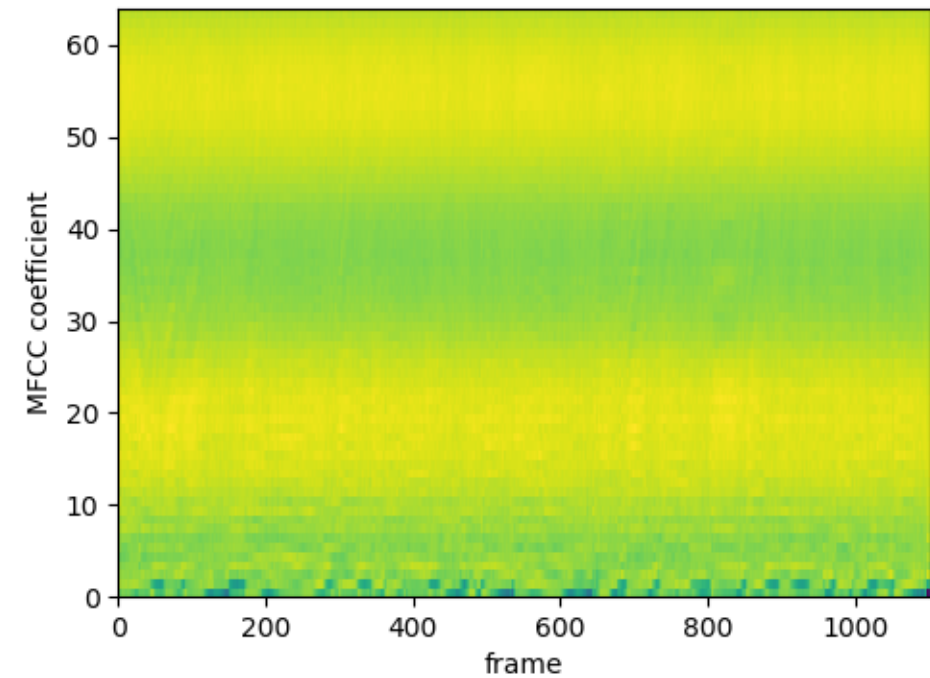
# Demo(2): Effect of num Mel-Filter banks

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**12 banks MFCC**



**64 banks MFCC**

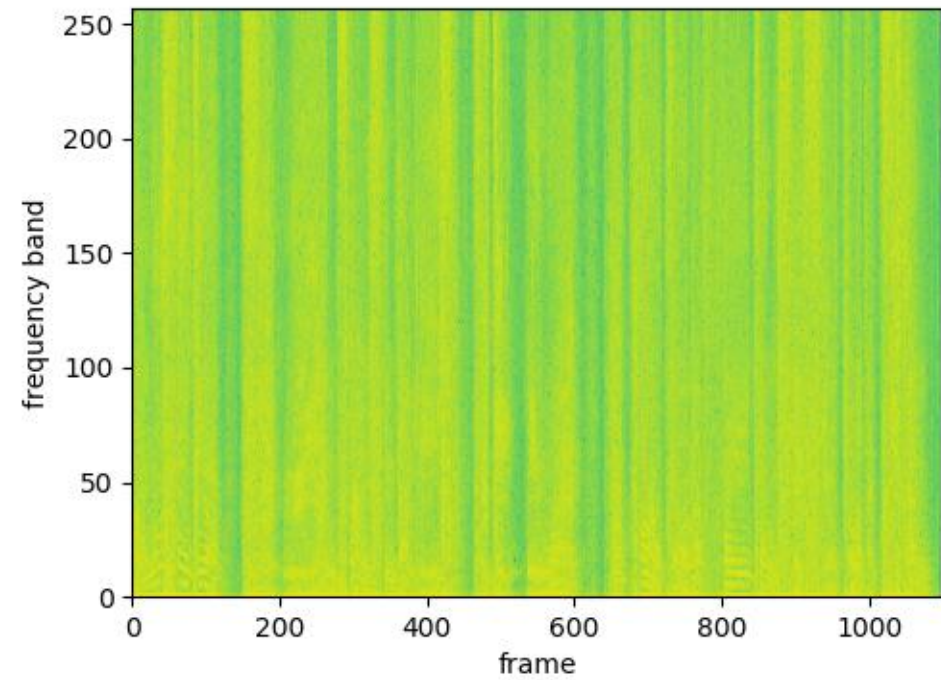
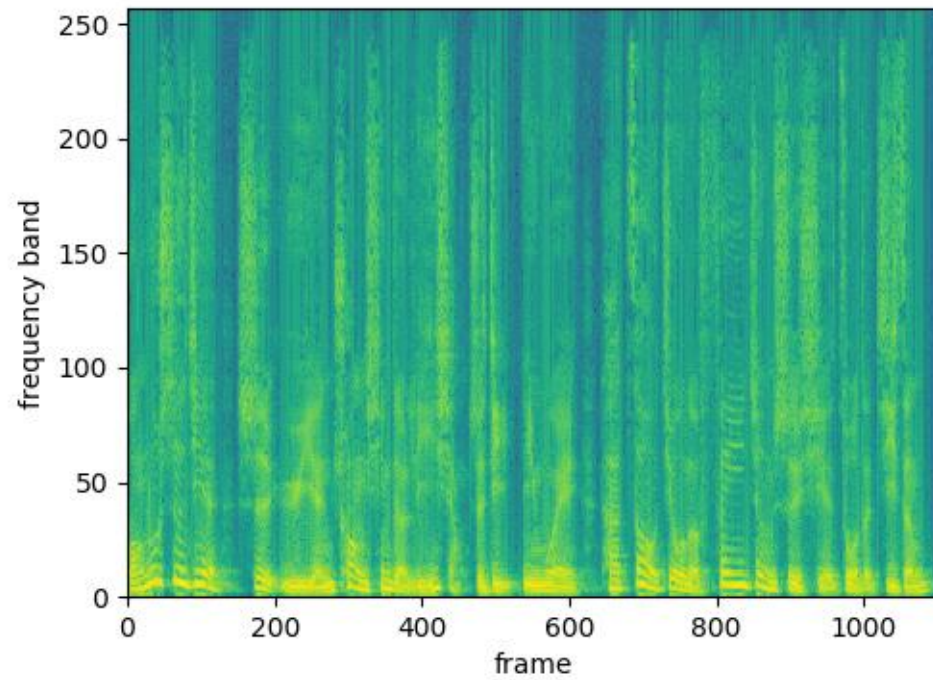


# Demo(3): Ori vs Reconstructed

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**64 banks, magnitude is log-scaled**

Original signal vs. Reconstructed signal





# Demo(4): Reconstructed Audio

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- 64 banks pre-emphasized: 
- 64 banks NOT pre-emphasized: 
- 12 banks pre-emphasized: 
- 12 banks NOT pre-emphasized: 
- Original: 

# Report questions:

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1. **Question 1:** What are the artifacts and distortions in the reconstructed audio? Suggest what the causes of these degradations are. (i.e. which sections of the MFCC extraction process are not invertible?)
2. **Question 2:** Experiment with different frame length, step length, and number of fbanks; discuss what effects each of them has in the reconstruction process.
3. **(Bonus 1)** Aside from setting optimal parameters, what can be added in the reconstruction algorithm to improve the end quality? Implement your proposal and present some experiments of it.
4. **(Bonus 2)** We did not perform dimension reduction/reconstruction in the DCT/inv-DCT sections. Modify those parts such that we have a complete algorithm that performs compression/decompression. Discuss how this influences the reconstruction quality.